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Signature of the Controller of Examination

Signature of the Student with date

Signature of the Invigilator with date

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INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
 - i. Writing their Hall Ticket No. and name in any part of the answer booklet.
 - ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
 - iii. Writing religious symbols.
 - iv. Either seeking or providing any assistance to the fellow students in the examinations.
 - v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
 - vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
 - vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**



SPACE FOR ROUGH WORK



Unit - I

2)

a) Given data :

$$f(x) = e^{-x}$$

intervals of $(0, 2\pi)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \quad \text{--- (1)}$$

$$a_0 = \frac{2}{\pi} \int_0^{2\pi} f(x) dx.$$

$$a_n = \frac{2}{\pi} \int_0^{2\pi} f(x) \cos nx dx.$$

$$b_n = \frac{2}{\pi} \int_0^{2\pi} f(x) \sin nx dx.$$

$$a_0 = \frac{2}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_0 = \frac{2}{\pi} \int_0^{2\pi} e^{-x} dx.$$

$$a_0 = \frac{2}{\pi} \left[e^{-x} \right]_0^{2\pi} dx$$

$$a_0 = \frac{2}{\pi} [e^{-2\pi} - e^0]$$



$$a_0 = \frac{2}{\pi} \left[e^{2\pi} \right]$$

$$a_0 = \left(\frac{e^{2\pi}}{\pi} \right)$$

$$a_n = \frac{2}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{2\pi} e^{-x} \cos nx \, dx.$$

$u = e^{-x}$	$v = \cos nx \, dx$
$u' = -e^{-x}$	$\rightarrow v_1 = \frac{\sin nx}{n}$
	$\rightarrow v_2 = -\frac{\cos nx}{n^2}$

$$\Rightarrow \frac{2}{\pi} \int_0^{2\pi} e^{-x}$$

$$\Rightarrow \frac{2}{\pi} \left[e^{-x} \frac{\sin nx}{n} + \frac{e^{-x} \cos nx}{n} \right]_0^{2\pi}$$

$$\Rightarrow$$

$$\Rightarrow \frac{2}{\pi} \left[e^{2\pi} \left(\frac{\sin 2\pi n}{n} \right) + \frac{e^{2\pi} \cos 2\pi n}{n^2} - e^0 \frac{\sin 0}{n} + e^0 \frac{\cos 0}{n^2} \right]$$

we know that

$\sin 2\pi n = 0$	$\cos 2\pi n = 1$
$\sin 0 = 0$	$\cos 0 = 1$



From these.

$$\Rightarrow \frac{2}{\pi} \left[\frac{e^{2\pi} + 1}{n^2} - 0 \right]$$

$$\Rightarrow -\frac{2e^{2\pi}}{\pi}$$

$$\Rightarrow \boxed{a_n = -\frac{2e^{2\pi}}{\pi}}$$

$$b_n = \frac{2}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx.$$

$$b_n = \frac{2}{\pi} \int_0^{2\pi} e^{-x} \sin nx \, dx.$$

~~$b_n = \frac{2}{\pi}$~~ $\Rightarrow$ By Applying uv = rule

$$\boxed{uv = u_1 v_1 + v_1 v_2}$$

$u = e^{-x}$	$v = \sin nx \, dx.$
$u_1 = e^x$	$v_1 = \frac{-\cos nx}{nx}$
	$v_2 = \frac{-\sin nx}{n^2}$

$$b_n = \frac{2}{\pi} \left[e^{-x} \frac{-\cos nx}{n} + e^x \frac{\sin nx}{n^2} \right]_0^{2\pi}$$



$$\Rightarrow b_n = \frac{2}{\pi} \left[\int_0^{-2\pi} \frac{-\cos 2n\pi}{n} + \frac{e^{-2\pi} - \sin n\pi}{n^2} \right] - \left(e^0 - \frac{\cos 2n\pi}{n} + \frac{e^0 - \sin n\pi}{n^2} \right)$$

$$\frac{e^0 (-\sin n\pi)}{n^2}$$

$$b_n = \frac{2}{\pi} \left[\left(2 \left(-\frac{1}{n} \right) + 0 - 0 \right) \right]$$

$$b_n = \frac{2}{\pi} \left[-\frac{2}{n} \right]$$

$$b_n = \frac{-4}{\pi n}$$

substitute the a_0 , a_n & b_n values, in equation ①.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \quad dx,$$

$$\text{for } a_0 = \frac{e\pi}{\pi}$$

$$a_n = \frac{-e\pi}{\pi}$$

$$b_n = \frac{2}{\pi n}$$



$$f(x) = \frac{\left(\frac{c\pi}{\pi}\right)}{2} + \sum_{n=1}^{\infty} \frac{-c\pi}{\pi} \cos nx + \frac{2\pi}{\pi n} \sin nx \, dx$$

2)

b) Given data :

Half range cosine series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (1)}$$

$$f(x) = x$$

intervals of $(0, 2)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$a_0 = \frac{2}{\pi} \int_0^2 f(x) \, dx$$

$$a_0 = \frac{2}{\pi} \int_0^2 x \, dx$$



$$a_0 = \frac{2}{\pi} \int_0^2 f(x) dx$$

$$a_0 = \frac{2}{\pi} \int_0^2 \left(\frac{x^{1+1}}{1+1} \right) dx$$

$$a_0 = \frac{2}{\pi} \int_0^2 \frac{x^2}{2} dx$$

$$a_0 = \frac{2}{\pi} \left[\frac{x^3}{3} - \frac{0^3}{3} \right]$$

$$a_0 = \frac{2}{\pi} \left[\frac{8}{3} \right]$$

$$\boxed{a_0 = \frac{8}{3\pi}}$$

$$a_n = \frac{2}{\pi} \int_0^2 f(x) \cos nx dx$$

$$a_n = \frac{2}{\pi} \int_0^2 x \cos nx dx$$

By Applying Bernoulli's rule

$$\boxed{uv_1 = uv_1 + v_1' u_1}$$



$$u = x$$

$$v' = \cos x$$

$$u' = 1$$

$$v = \left(\frac{\sin x}{n} \right)$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \frac{\cos' \sin x}{n} dx$$

$$\frac{2}{\pi} \left[x \frac{\sin 2n}{n} - \left(0 - \frac{\sin 2n}{n} \right) \right]$$

$$\frac{2}{\pi} \left[x \left(\frac{\sin 2n}{n} \right) \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{\sin 2n}{n} \right]$$

$$a_n = \frac{1}{\pi} (\sin 2)$$

$$a_n = \left(\frac{\sin 2}{\pi} \right)$$

By substituting the value's in equation ①

$$a_0 = \frac{8}{2\pi}$$

$$a_n = \left(\frac{2 \sin 2}{\pi} \right)$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \, dx = \frac{8}{2\pi} + \sum_{n=1}^{\infty} \left(\frac{2 \sin}{\pi} \right) \cos nx \, dx = 0 =$$



unit - IV

2)

a)

Given data

$$\mathcal{L}^{-1} \frac{s}{(s^2 + a^2)^2}$$

$$\mathcal{L}^{-1} \left(\frac{s}{s^2 + a^2} \right) \quad \mathcal{L}^{-1} \left(\frac{1}{(s^2 + a^2)} \right)$$

$$\mathcal{L}^{-1} (\cos a) \quad \mathcal{L}^{-1} (\sin a)$$

(using convolution theorem)

$$\bar{f}(s) = \bar{g}(s) = (\bar{f}(u) * \bar{g}(t-u)) du$$

$$f(s) = (\cos(a) * \sin(t-u))$$

$$\frac{2}{\pi} \int_0^t \cos(a) \sin(t-u) du$$

$$2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$\Rightarrow \frac{2}{\pi} \int_0^t \sin u(t+u) - \sin(u-t-u) du$$



$$\Rightarrow \frac{2}{1} \int_0^t \sin(2v) t - \sin(2v) t \, dv$$

$$\Rightarrow \frac{2}{1} \int_0^t (\sin 2v - \sin 2v) \, dv$$

$$\Rightarrow \frac{2}{1} \left[\frac{-\cos(v+t-v)}{(v+t-v)} - \frac{-\cos(v+t-v)}{(v-t-v)} \right]_0^t$$

$$\Rightarrow \frac{2}{1} [-\cos t - \cos 0]$$

$$\Rightarrow \frac{2}{1} (-\cos t)$$

$$\Rightarrow \left(\frac{2 \cos t}{1} \right) = \frac{s}{(s^2 + a^2)^2}$$

8)

b) Given data:

laplace transform's

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 3y = e^{-t}$$

$$y(0) = y'(0) = 1$$



By taking equation into

$$y'' + 4y' + 3y = e^{-t}$$

formula:

$$y'' = s^2(y) + s y(0) - y'(0)$$

$$y' = s(y) - y(0)$$

$$\rightarrow y'' + 4y' + 3y = e^{-t}$$

$$\rightarrow s^2(y) + s y(0) + y'(0) + s(y) - y(0) + 3y(0) = e^{-t}$$

$\rightarrow$ By the given notation $y(0) = 0$ $y'(0) = 1$

$$\rightarrow s^2 + 0 + 1 + s + 0 + 0$$

$\rightarrow$ By Arranging the values

$$\rightarrow s^2 + s + 1 = e^{-t}$$

By the $\left(\frac{b^2 \pm \sqrt{4Ac}}{2A} \right)$

we get $(s+1) (s+2)$



$$\frac{e^{-t}}{(s+1)(s+1)(s-2)} = \frac{A}{s+1} + \frac{B}{(s+1)} + \frac{C}{(s-2)}$$

$$\frac{e^{-t}}{(s+1)(s+1)(s-2)} = \frac{A(s+1)(s-2) + B(s+1)(s-2) + C(s+1)(s-1)}{(s+1)(s+1)(s-2)}$$

$$\Rightarrow e^{-t} = A(s+1)(s-2) + B(s+1)(s-2) + C(s+1)(s-1)$$

$$\boxed{\text{put } s = -1}$$

$$A(s+1)(s-2)$$

$$A(-1)(-2)$$

$$A(-2)(-3)$$

$$\boxed{A = 6}$$

$$\text{put } s = -1$$

$$B(s+1)(s-2)$$

$$B(2)(-1)$$

$$\boxed{B = -2}$$

$$\text{put } s = s-2$$

$$C(s+1)(s-1)$$

$$C(2+1)(2-1)$$

$$C(3)(1)$$

$$\boxed{C = 3}$$

$$\frac{e^{-t}}{(s+1)(s+1)(s-2)} = \frac{6}{(s+1)} + \frac{-2}{(s+1)} + \frac{3}{(s-2)}$$

$$y^2 + 4y + 3y = e^{-t} = \frac{e^{-t}}{s+1(s+1)(s-2)} = \frac{6}{s+1} + \frac{-2}{(s+1)} + \frac{3}{s-2}$$



unit - V

9)

Given data:

~~$$\frac{dv}{dx} + 2 \frac{dv}{dy} + v$$~~

10) Given data

$$a) \quad \frac{dv}{dx} = 4 \frac{dv}{dy}$$

$$v(x, y) = 8e^{-3y}$$

separation of variables

$$\frac{dv}{dx} = 4 \frac{dv}{dy} = 8e^{-3y}$$

$$xx' = xy' \quad xy'$$

$$x'xy = xy' = 0xy$$

$$\frac{x'y}{xy} = \frac{xy'}{xy} = \frac{xy}{xy}$$

$$\Rightarrow \frac{x'}{x} = \frac{y'}{y} = 0$$



$$\text{assumo } \frac{x'}{x} = \frac{y'}{y} = k$$

$$-(k+1) \frac{x'}{x} = \frac{y'}{y} = k$$

$$\cancel{(k+1)} = \cancel{\frac{y'}{y}}$$

$$x' - y' = k(xy)$$

$$\cancel{x - y}$$

$$\log e = k \log(xy)$$

$$\log e = k \log xy$$

$$e = k(xy)$$

$$e^{-3y} = k$$

$$\frac{dy}{dx} = \frac{dv}{dy} = 8e^{-3y}$$



Unit: III

3)

a)

Given data:

$$L \left\{ \int_0^t e^{-t} \cos t \, dt \right\}$$

In this first shifting theory

$$L \int_0^t \cos t \, dt = \int_0^t \frac{\sin t}{0.2}$$

$$= \int_0^t \left(\frac{\sin t}{2} \right)$$

$$\cancel{L \int_0^t \cos t \, dt} = L \left(\frac{\sin t}{2} \right)_0^t$$

$$= L \left[\frac{\sin t}{2} \right]_0^t$$

$$L \frac{1}{2} [\sin t - \sin 0]$$

$$L \frac{1}{2} [\sin t]$$



By Applying the first shifting theorem.

$$\mathcal{L} \left\{ \int_0^t e^{-t} \cos t \, dt \right\}$$

$$\boxed{e^{-t} \quad s = s+a}$$

$$\mathcal{L} = \frac{1}{2} \left[\sin t \right] \quad \text{By integrating.}$$

$$\mathcal{L} = \frac{1}{2} \left[\frac{1}{s^2 + \cancel{0}^2} \right] = \frac{1}{2} (\sin t)$$

$$\mathcal{L} = \frac{1}{2} \left[\frac{1}{s^2 + t^2} \right]$$

By Applying th (S-1) to s.

$$\frac{1}{2} \left[\frac{1}{(s-1) + t^2} \right]$$

$$\mathcal{L} \left\{ \int_0^t e^{-t} \cos t \, dt \right\} = \frac{1}{2} \left(\frac{1}{(s-1) + t^2} \right)$$



Unit II

3)

b) Given data:

$$f(x) = \begin{cases} 1 & \text{if } 0 < x < 2 \\ 0 & \text{if } x > 2 \end{cases}$$

Fourier sin series and cosh series
sin series.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{1}{2} \int_0^2 f(x) \sin nx \, dx = \frac{1}{2} \int_0^2 (1) \sin nx \, dx$$

$$b_n = \frac{1}{2} \int_0^2 (1) \sin nx \, dx = \frac{1}{2} \left[\frac{-\cos nx}{n} \right]_0^2$$

$$b_n = \frac{1}{2} \left[(1) \left(\frac{-\cos nx}{n} \right) \right]_0^2 \quad \frac{0 \cdot x - \cos nx}{n} = 0$$

$$b_n = \frac{1}{2} \left[(1) \left(\frac{-\cos nx}{n} \right) - \left((1) \left(\frac{-\cos n \cdot 0}{n} \right) \right) \right]$$

$$b_n = \frac{1}{2} \left[\left(\frac{-\cos n \cdot 2}{n} \right) - \left(\frac{-1}{n} \right) \right]$$

$$\frac{1}{2n^2} [2 \cos n - 1]$$



$$b_n = \frac{2}{2n^2} (2 \cosh n - 1)$$

$$b_n = \frac{2 \cosh n - 1}{2n^2}$$

$$b_n = \frac{\cosh n - 1}{n^2}$$

Fourier cosh series.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cosh nx$$

$$a_0 = \frac{1}{\pi} \int_0^2 f(x) dx$$

$$a_0 = \frac{1}{\pi} \int_0^2 (1) dx$$

$$\frac{1}{\pi} \int_0^2 0 dx$$

$$a_0 = \frac{1}{\pi} \int_0^2 x dx$$

$$a_0 = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^2$$

$$a_0 = \frac{1}{\pi} \left[\frac{4}{2} \right]$$

$$a_0 = \frac{1}{\pi} (2)$$

$$a_0 = \left(\frac{2}{\pi} \right)$$



$$a_n = \frac{1}{\pi} \int_0^2 f(x) \cos nx \, dx.$$

$$a_n = \frac{1}{\pi} \int_0^2 1 \cos nx \, dx.$$

$$a_n = \frac{1}{\pi} \left[\frac{\sin nx}{n} \right]_0^2$$

$$a_n = \frac{1}{\pi} \left[\frac{\sin 2n}{n} - 0 \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{\sin 2n}{n} \right]$$

$$a_n = \left(\frac{2 \sin n}{n} \right)$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$\left(\frac{\frac{2}{\pi}}{2} \right) + \sum_{n=1}^{\infty} \left(\frac{2 \sin n}{n} \right) \cos nx \, dx, \quad \underline{\underline{= 0}}$$



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